FINAL REPORT

Supply Chain Delay Analysis

Why are so many orders late, and what is it costing?

Power BI dashboard on 47,396 orders (2022 to 2024)

47,396	13.7%	53.5%	32.8%	Rs 142.7M
Orders analysed	Arrive on time	Delayed	Cancelled	Estimated delay cost

The short version

I took a messy file of 50,500 supply chain orders, cleaned it, and built a one-page Power BI dashboard to find out where deliveries go wrong. The answer is uncomfortable: **only about 1 in 7 orders arrives on time.** More than half are delayed and a third are cancelled.

Two causes stand out. **Late suppliers** and **warehouse congestion** account for 67% of the orders that have a recorded reason. Everything else (weather, strikes, customs, traffic and so on) is small by comparison.

> **What I would tell the business:** fix supplier delivery dates and warehouse congestion first. Together they are two thirds of the problem, and both are things the company can control.

Who this report is for

Anyone who wants the result without opening the dashboard: a recruiter, a hiring manager, or a manager in operations. Each section starts with the finding, then shows the number behind it.

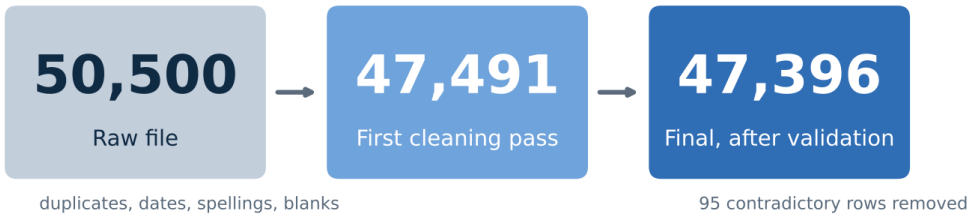
1. The problem

A company shipping about 47,000 orders over three years kept missing its delivery dates. Management wanted three answers:

Question	Where the answer is
How bad is the delay problem?	Section 3, headline numbers
Where do the delays come from?	Section 4, causes, stages and suppliers
What is it costing, and what should we do?	Section 4 and Section 5

2. The data and how I cleaned it

The raw file had 50,500 rows and a lot of mess: repeated order IDs, dates written in different formats, the same status spelled in several ways, and blank cells. I cleaned it in Excel, then loaded it into Power BI.

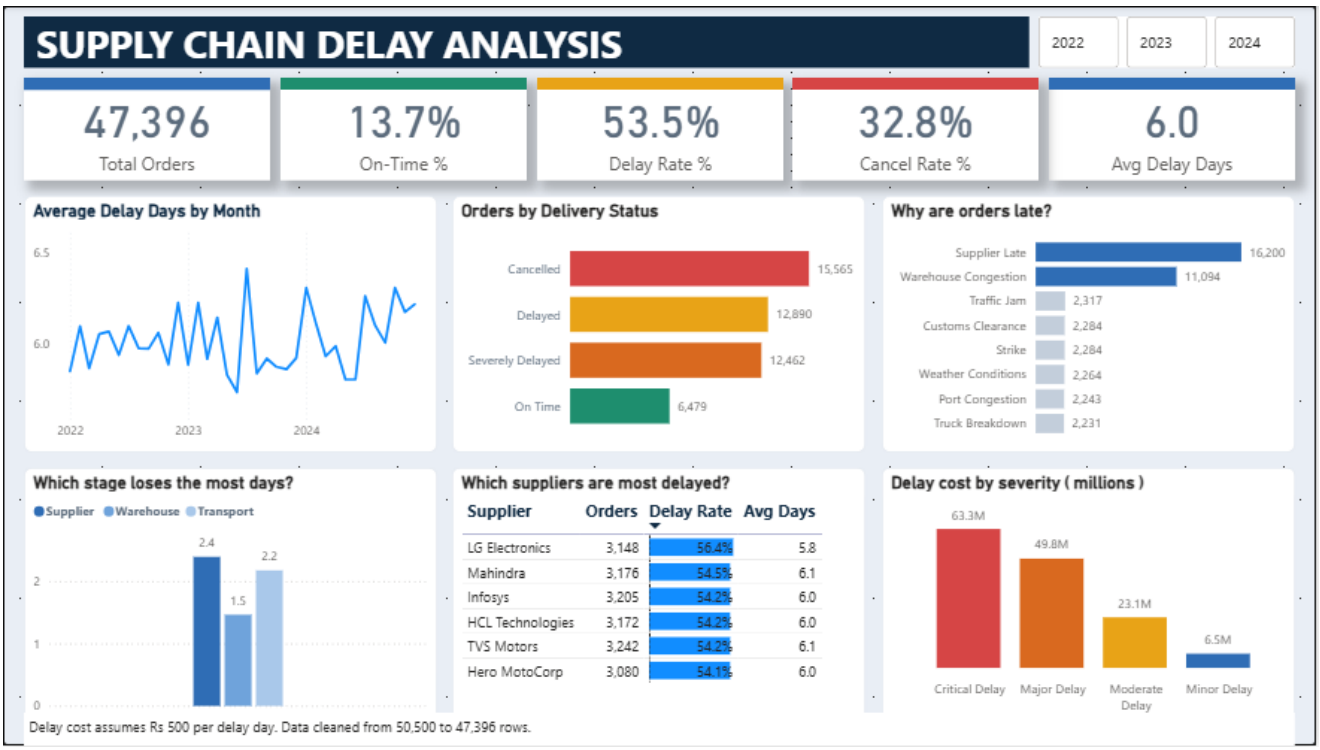


Cleaning step	What I did
Duplicates	Removed repeated order IDs
Dates	Converted mixed date formats into one
Spellings	Made status, reason and supplier names consistent
Blanks	Dealt with missing values column by column
Validation	Power BI removed 95 rows whose delay figures contradicted each other
Result	**47,396 clean orders** (from 50,500)

The cleaned file has 47,491 rows. The dashboard shows 47,396 because of the 95 contradictory rows in the validation step.

3. The dashboard and the headline numbers

Everything fits on one page so a manager can read it in under a minute.



The Power BI dashboard. Year buttons at the top right filter every visual.

Headline numbers

	Measure	Value	What it means
	On time	13.7% (6,479 orders)	Arrived by the promised date
	Delayed	27.2% (12,890)	Late, but not severely
	Severely delayed	26.3% (12,462)	Very late
	Cancelled	32.8% (15,565)	Never delivered
	Average delay	6.0 days	Across delayed orders
	Total delay cost	Rs 142.7M	Estimated, Rs 500 per delay day

The Delay Rate on the dashboard (53.5%) is delayed plus severely delayed orders divided by all orders.

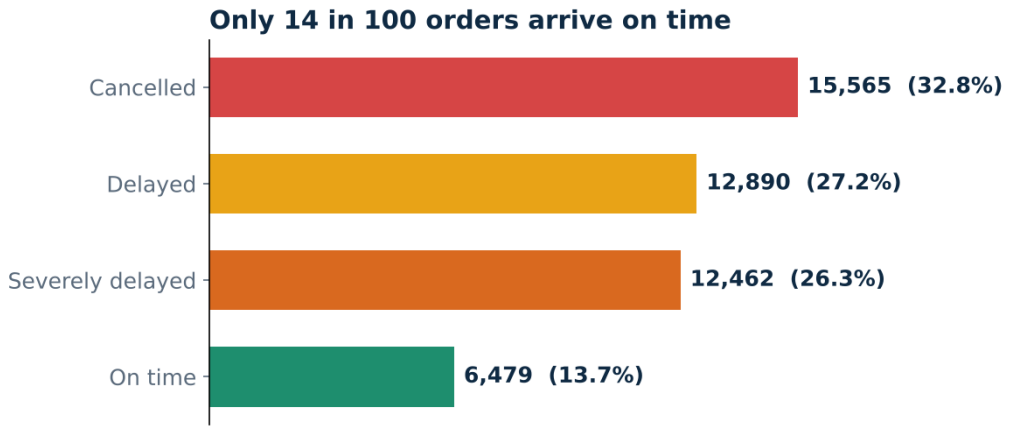
How the data is set up

The model is deliberately simple: one date table connected to the orders table. That is why the Year buttons filter every chart at once.

Table	Role	Link
Dim_Date	Calendar: year, month, quarter	One date to many orders
Fact_Orders	One row per order: status, reason, delays, cost	Single direction, so the Year slicer filters every chart

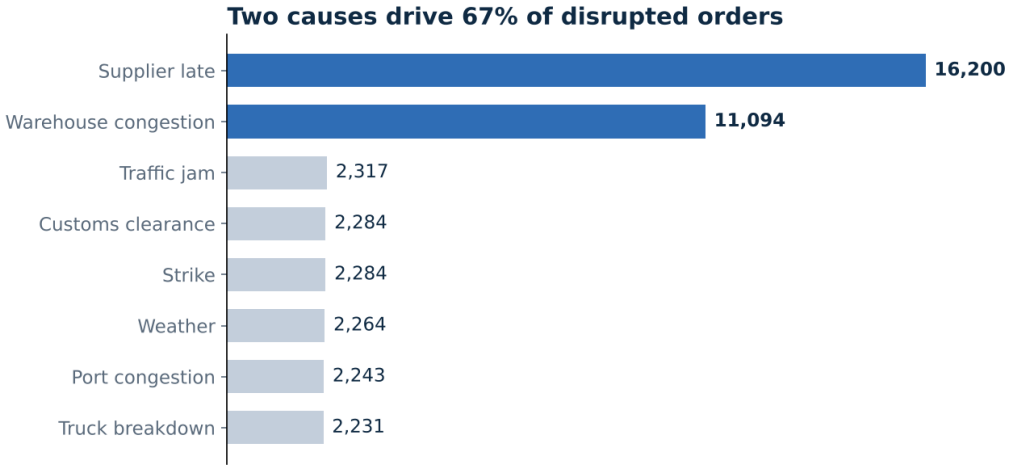
4. What I found

Finding 1: Only 14 in 100 orders arrive on time



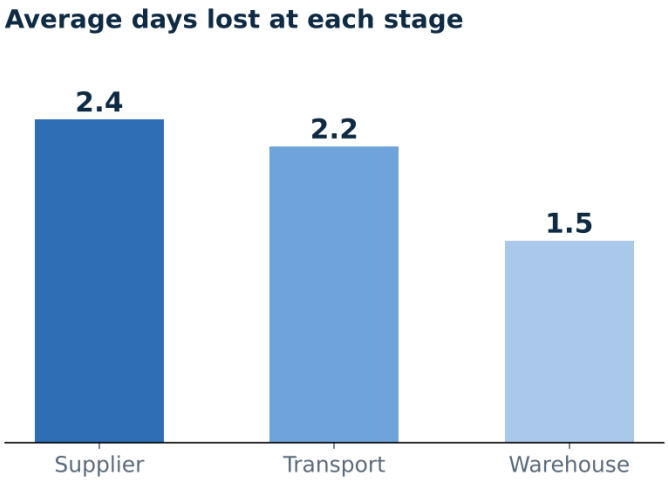
Finding 2: Two causes drive two thirds of the problem

Supplier late (16,200 orders) and warehouse congestion (11,094) together make up 67% of orders with a recorded reason. The other six causes are each close to 2,300.



Finding 3: Suppliers lose the most time, and it is not one bad supplier

A delayed order loses on average **2.4 days at the supplier**, 2.2 in transport and 1.5 in the warehouse.



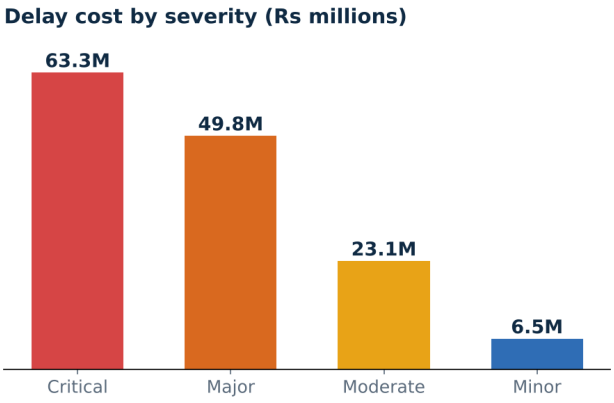
The six most delayed suppliers are very close to each other, so the problem is spread across the supplier base:

Supplier	Orders	Delay rate	Avg delay (days)
LG Electronics	3,148	56.4%	5.8
Mahindra	3,176	54.5%	6.1
Infosys	3,205	54.2%	6.0
HCL Technologies	3,172	54.2%	6.0
TVS Motors	3,242	54.2%	6.1
Hero MotoCorp	3,080	54.1%	6.0

The gap between the worst and the sixth worst is only 2.3 points.

Finding 4: Most of the money is in the worst delays

Critical and major delays cost Rs 113.1M of the Rs 142.7M, about 79%. Minor delays cost just Rs 6.5M.



5. What I would recommend

Priority	Action	Why
1	Set firm delivery dates with every supplier and track them	Supplier delays are the biggest cause (16,200 orders) and no single supplier stands out
2	Reduce warehouse congestion	Second biggest cause (11,094 orders) and fully inside the company's control
3	Put the effort into critical and major delays	They hold about 79% of the Rs 142.7M cost
4	Study cancellations separately	At 32.8% they are the biggest group, but the data does not say why

6. What this analysis cannot tell you

I would rather be upfront about the limits:

• **Delay cost is an estimate.** It assumes a flat Rs 500 per delay day, which is my assumption, not a number from the company's accounts.

• **Cancellation reasons are missing**, so I can say how many orders were cancelled but not why.

• **The supplier table shows only the six highest delay rates.**

• **It describes the past.** I did not build a forecast.

7. How I built it

Step	Tool	What I did
Clean	Excel	Removed duplicates, fixed dates and spellings, handled blanks
Load	Power Query	Loaded the data and set the data types
Model	Power BI	Date table linked to the orders table, one-to-many
Measure	DAX	Total orders, on-time %, delay rate, cancel rate, average delay, total delay cost
Present	Power BI	One-page dashboard: KPI cards, six charts, Year slicer

What I learned

Most of the work was cleaning, not charting. One wrong KPI, for example a delay rate calculated on the wrong row count, would have changed the whole story, so I checked every number on the page back to the source data.

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